STREAMING DATA & OPERATIONS · CASE STUDY

24.7% of incidents breach the SLA. Schema changes are the biggest lever.

End-to-end technical and business case on 160,527 feed-days and 5,485 incidents (synthetic data anchored on public Simply.TV milestones).

5,485
incidents, Jun 2023–Sep 2026

24.7%
outside SLA (1,337 of 5,404)

37%
of breaches come from schema drift (22% of incidents)

316
bad deliveries that never opened a ticket

13 h vs 0.2 h
detection: client vs automatic

1 PROBLEM
Median fix 19 h, but the slowest 10% take over 94 h.

2 CAUSE
Schema changes break SLA 41% of the time; North America 31.8% vs DACH 19.8%.

3 ACTION
Alert on schema change and completeness drops; flag the riskiest 20% of tickets at opening.

4 NEXT STEP
90-day pilot on real data with a control group.

Technical work

- Python ETL to a SQLite star schema; 18 SQL queries (window functions, anti-joins)
- NLP triage: 84.7% accuracy, macro F1 0.835
- SLA-risk model: AUC 0.726 vs 0.690 baseline; bootstrap interval excludes zero; wins in 4/4 time windows; calibration error 1.6%
- Weekly forecast beats naive (MAE 4.51 vs 5.84); anomaly detection for silent bad days
- Rule-based incident investigator: writes the ticket and the client message
- Local FastAPI, 19 tests, CI, security review

Business view

- Flagging the riskiest 20% captures about 40% of breaches
- With stated assumptions (30% faster handling, EUR 500 per breach): about 87 breaches avoided per year. The cost per breach is an assumption to replace with real data.
- 30/60/90 plan: connect real logs, run alerts in shadow mode, pilot with a control group
- Honest limits: synthetic data; investigator family accuracy 53% vs 45% baseline

Live dashboard: christianmg22.github.io/simplytv-streaming-ops · Code: github.com/ChristianMg22/simplytv-streaming-ops · Christian Mercado